Artificial Intelligence and the Economy: Productivity, Macroeconomic Adjustment, Financial Stability, and Policy Challenges

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1 Literature Review

Artificial intelligence changes the cost of prediction and the productivity of cognitive tasks (Agrawal, Gans, & Goldfarb, 2018). Recent generative AI systems differ from earlier automation technologies: rather than replacing routine physical or clerical labor, they assist decision-making across occupations that were previously insulated from automation (Eloundou, Manning, Mishkin, & Rock, 2024). Organizational AI adoption rose from 55 to 78 percent in a single year, and private investment in generative AI reached $33.9 billion in 2024 (Stanford Institute for Human-Centered Artificial Intelligence, 2025). As Brynjolfsson, Korinek, and Agrawal (2025) argue, the economics literature has moved from documenting task-level productivity gains to asking whether AI will reshape aggregate growth and labor-market inequality.

1.1 Micro productivity and labor-market exposure

Randomized experiments credibly identify AI's causal effect on worker productivity. Noy and Zhang (2023) run a preregistered experiment with 453 professionals and find that ChatGPT access reduces writing-task completion time by 40 percent and raises output quality by 18 percent, with the largest gains accruing to lower-performing workers. Noy and Zhang randomize assignment and blind-evaluate output, but external validity is limited: the sample uses a single online platform and a single task type. They do not test whether these effects persist in routine work or scale to other cognitive domains.

Workplace data confirm these experimental results at scale. Brynjolfsson, Li, and Raymond (2025) study 5,172 customer support agents at a Fortune 500 firm and report average productivity gains of 14 percent. The bottom skill quintile improves by 34 percent while top-quintile workers gain almost nothing. The AI encodes tacit practices of the best agents and transfers them to novices; the within-firm skill distribution compresses. The staggered rollout provides within-firm variation, though the authors do not address whether early adopters differ systematically from late adopters in ways that bias the estimates.

Roughly 80 percent of U.S. workers have at least 10 percent of their tasks exposed to large language models; about 19 percent are exposed on half or more (Eloundou et al., 2024). With complementary software, 47 to 56 percent of all tasks could be affected. Felten, Raj, and Seamans (2023) use a different methodology and find that higher-wage, higher-education occupations are most exposed, reversing the pattern from earlier automation. Both approaches measure technical potential, not realized adoption. The gap between exposure and adoption is the central measurement problem in this literature.

Physical automation carries a cautionary record. Between 1990 and 2007, one additional industrial robot per thousand workers reduced local employment by 0.2 percentage points and wages by 0.42 percent in U.S. commuting zones; displacement dominated reinstatement (Acemoglu & Restrepo, 2020). Whether AI follows this substitution pattern remains open. Autor (2024) argues that AI could rebuild middle-class expertise in healthcare and software if deployed to support decision-making rather than to replace workers. But rebuilding expertise requires institutional choices that steer deployment toward complementarity, not cost reduction alone.

The task-level evidence is internally consistent: generative AI raises productivity in cognitive tasks, with the largest gains at the bottom of the skill distribution. Whether these effects aggregate into GDP gains above a few tenths of a percent per year is a separate question. Aggregation requires that firms adopt the technology and invest in complementary organizational redesign, both uncertain.

1.2 Macroeconomic adjustment: growth, output, and inflation

Macro estimates of AI's effect on total factor productivity differ by an order of magnitude. The disagreement is less about model structure than about calibration, and it reflects genuine uncertainty about adoption.

Acemoglu (2024) provides the cautious benchmark. Applying Hulten's theorem to a task-based framework, he estimates that AI will raise TFP by 0.53 to 0.66 percent over ten years, roughly 0.06 percent per year. The logic is arithmetic: about 20 percent of tasks are exposed (Eloundou et al., 2024), 23 percent of those can be profitably automated within a decade, and per-task cost savings average 27 percent. The estimate rests on a specific assumption: Hulten's theorem requires that marginal cost shares equal output elasticities. This holds under competitive conditions but may break down if AI introduces non-marginal technology shifts or if market power in AI provision is substantial. Aghion and Bunel (2024) use the same decomposition but argue for broader exposure (60 percent of tasks rather than 20) and higher per-task gains averaging 40 percent. Their baseline is 0.68 percentage points of additional annual TFP growth, ten times Acemoglu's figure, with a range of 0.07 to 1.24. The 20 percent exposure figure comes from conservative task-level rubrics that exclude complementary software; the 60 percent figure includes it. Which is correct depends on whether firms build applications on top of foundation models, an adoption question that exposure measures alone cannot answer.

General equilibrium forces can amplify or dampen the translation from task productivity to aggregate output. Aldasoro, Doerr, Gambacorta, and Rees (2024) build a multi-sector model in which AI enters as a sector-specific productivity shock calibrated to an industry-level exposure index. AI could raise annual productivity growth by about one percentage point, but inflation effects depend on expectations: unanticipated gains are disinflationary, while anticipated gains trigger consumption and investment surges that generate inflationary pressure. The model also reveals that a sector's initial AI exposure has little correlation with its long-run output gain, because input-output linkages redistribute productivity effects across the economy.

Cross-country evidence adds a distributional dimension. Gambacorta, Kharroubi, Mehrotra, and Oliviero (2026) study 56 economies using a Rajan-Zingales identification strategy and find that generative AI benefits advanced economies more than emerging markets in the short run. The gap reflects sectoral composition and digital-infrastructure deficits. AI adoption may therefore widen rather than narrow global income gaps.

Baumol's cost disease may constrain AI's long-run growth effect. Aghion, Jones, and Jones (2018) model AI as task automation within a CES framework and show that automated sectors become cheap and lose GDP weight, while bottleneck sectors determine the long-run growth rate. A growth explosion requires full automation of goods production or automation of idea production itself, neither of which current AI systems accomplish. Why, then, has measured productivity not yet responded? Brynjolfsson, Rock, and Syverson (2018) argue that implementation lags matter most: AI requires complementary intangible investment in organizational redesign that national accounts do not capture. Brynjolfsson, Rock, and Syverson (2021) formalize this as a J-curve: early in adoption, resources devoted to intangibles depress measured productivity; later, the accumulated stock generates output without being credited as input. By 2017, their estimates suggest true TFP exceeded official measures by 15.9 percent because of unmeasured intangible capital.

Task-level productivity gains do not translate mechanically into aggregate output. The translation depends on adoption costs and complementary investment, both slow and hard to measure.

1.3 AI in finance and central banking

AI in finance presents a trade-off between information-processing gains and systemic risk. Adoption is faster here than in most industries: over 84 percent of surveyed institutions already use AI, and 86 percent plan to expand generative AI applications (Aldasoro, Gambacorta, Korinek, Shreeti, & Stein, 2024). Algorithmic trading is roughly 70 percent of U.S. equity volume, and AI analysis of FOMC announcements now moves prices within seconds (International Monetary Fund, 2024). These tools improve credit scoring and central-bank nowcasting (Bank for International Settlements, 2024), but speed and scale also create conditions for correlated behavior at systemic scale.

Herding is the central systemic risk. Institutions face similar regulatory constraints and train on the same public data, so they converge on correlated strategies even without explicit coordination. When algorithms simultaneously detect the same adverse signal, they withdraw liquidity at the same moment (Adrian, Mosk, & Wu, 2025; Financial Stability Board, 2024). During the March 2020 turmoil, AI-powered ETFs showed increased portfolio turnover precisely when markets were most stressed (International Monetary Fund, 2024). Adrian et al. (2025) formalize the mechanism: rising volatility triggers automated deleveraging; the resulting price declines raise volatility further. The key assumption is that risk-management algorithms share a common signal structure, so individually rational deleveraging becomes collectively destabilizing. Because reaction times compress to milliseconds, localized stress can escalate before human intervention.

Third-party concentration compounds these vulnerabilities. A small number of cloud providers and foundation-model developers supply infrastructure to much of the financial sector, and Financial Stability Board (2024) identify this concentration as one of four systemic risks. A disruption at a single provider could propagate simultaneously across institutions, a correlation channel that existing stress-testing frameworks do not capture. This concentration also matters for the macro debate: market power in AI provision may violate the competitive conditions that Hulten's theorem requires, widening the gap between task-level exposure and realized aggregate gains. AI changes markets faster than supervisory frameworks can adapt (International Monetary Fund, 2024).

1.4 AI as a research technology for economists

Korinek (2023) identifies six domains where large language models assist economists, including coding and mathematical derivation. He estimates productivity gains of 10 to 20 percent, primarily from automating routine cognitive subtasks while economists supply identification strategies and domain expertise. The main risk is hallucination: LLMs fabricate citations and produce confident but incorrect mathematical results. Human oversight remains indispensable.

A separate frontier involves converting unstructured data (text, images, satellite imagery) into structured variables for econometric analysis (Dell, 2025). Transfer learning allows fine-tuning on small domain-specific datasets without large computational resources. These methods may also help close the adoption-measurement gap identified in the macro literature: unstructured firm-level data on AI deployment could be converted into the structured evidence that calibrated models require. Reproducibility suffers because researchers rarely archive training data and model specifications, and data leakage between training and test sets can inflate reported accuracy (Dell, 2025; Korinek, 2023).

On the computational side, Fernández-Villaverde (2025) shows that deep neural networks can approximate policy and value functions in high-dimensional dynamic equilibrium models; heterogeneous-agent models with rich state spaces become feasible on standard hardware. Chi, Fan, Ghigliazza, Giannone, and Wang (2025) provide a counterpoint: well-regularized linear models perform nearly as well as deep networks for macroeconomic forecasting, suggesting limited marginal value of nonlinearity in this domain.

AI tools reduce certain research costs but do not replace identification or economic reasoning.

1.5 Policy challenges and open questions

The policy literature is less settled than the productivity literature. AI may raise aggregate welfare and worsen inequality simultaneously.

Korinek (2024) argues that AI threatens cognitive and creative occupations, not only routine work, so workers with specialized human capital may face larger income losses than blue-collar workers displaced by earlier automation. If gains accrue mainly to capital owners and firms controlling data and compute, the capital-labor distribution shifts in ways that erode tax bases built on labor income. The cross-country evidence reinforces this concern. Gambacorta et al. (2026) show that the advanced-economy advantage in AI-driven output carries distributional implications: the digital-infrastructure deficits that limit adoption in emerging markets also limit the policy tools available to compensate displaced workers.

Optimal robot-specific taxes are small. Thuemmel (2023) studies the question in a general equilibrium model with occupational choice and finds that the optimal tax or subsidy is on the order of one percent. At current prices, a modest subsidy is welfare-improving because it compresses the wage distribution from above. As robots become cheaper, the optimal policy switches to a tax, but the main welfare gains come from reforming income taxation, not from robot-specific instruments. The fiscal question that matters more is how to restructure income and capital taxation when labor's share of income is declining.

The literature largely ignores environmental costs. Bonfiglioli, Crinò, Filomena, and Gancia (2025) use a shift-share strategy across 722 U.S. commuting zones and find that localities with higher AI penetration experienced slower emissions declines between 2002 and 2022; without AI-driven growth, average emissions would have fallen approximately 37 percent more. The shift-share instrument uses pre-period industry composition interacted with national AI adoption trends; the result survives controls for local manufacturing decline, though Bonfiglioli et al. do not report pre-trend tests across exposure groups. Data centers require stable baseload power, and nearby plants shift generation toward fossil fuels. Training emissions for frontier models are growing rapidly (Stanford Institute for Human-Centered Artificial Intelligence, 2025).

The most pressing data need is firm-level adoption information linking AI use to productivity and intangible investment, because these data would directly discipline the parameter assumptions that drive the order-of-magnitude disagreement in macro estimates. Worker-level panel studies that track displacement and occupational mobility over longer horizons are also needed; the horizon matters because short-run augmentation and long-run substitution can coexist.

On the theoretical side, models that endogenize adoption costs rather than treating them as exogenous parameters would help explain why the same technology diffuses at different rates across sectors and countries. Multi-sector models should be calibrated with observed adoption costs rather than occupational exposure scores alone. In finance, transaction-level evidence on whether AI models generate correlated trading during stress episodes would give regulators a factual basis for recalibrating circuit breakers. Environmental research should combine data-center location with electricity-source variation to measure AI's carbon footprint with greater precision (Bonfiglioli et al., 2025; Brynjolfsson, Korinek, & Agrawal, 2025).

1.6 Concluding remarks

The economics literature on AI has advanced most on task-level productivity: experimental and workplace studies credibly show that generative AI raises output in cognitive tasks, with the largest gains at the bottom of the skill distribution. The central unresolved issue is aggregation. Macroeconomic models produce TFP estimates that differ by an order of magnitude because they disagree on the breadth of profitable automation and the required complementary investment. The productivity-paradox literature offers one reconciliation (measured gains will arrive with a lag), but whether AI raises long-run growth rates depends on whether it automates idea production, a possibility that growth theory takes seriously but empirical evidence has not tested. In finance, AI improves information processing while introducing herding and concentration risks that supervisors have not yet learned to measure. Policy remains underdeveloped on inequality and financial regulation. The most pressing need is not more micro experiments but better data on adoption and organizational adjustment, the missing links between task-level evidence and aggregate outcome.

References

Acemoglu, D. (2024). *The simple macroeconomics of AI* (Working Paper No. 32487). National Bureau of Economic Research.

Acemoglu, D., & Restrepo, P. (2020). Robots and jobs: Evidence from US labor markets. *Journal of Political Economy*, *128*(6), 2188–2244.

Adrian, T., Mosk, B., & Wu, J. (2025). *The future of AI in capital markets* (Discussion Paper No. 20748). Centre for Economic Policy Research.

Aghion, P., & Bunel, S. (2024). *AI and growth: Where do we stand?* (Policy Note). Federal Reserve Bank of San Francisco.

Aghion, P., Jones, B. F., & Jones, C. I. (2018). Artificial intelligence and economic growth. In A. Agrawal, J. Gans, & A. Goldfarb (Eds.), *The economics of artificial intelligence: An agenda.* Chicago: University of Chicago Press.

Agrawal, A., Gans, J., & Goldfarb, A. (2018). *Prediction machines: The simple economics of artificial intelligence*. Boston: Harvard Business Review Press.

Aldasoro, I., Doerr, S., Gambacorta, L., & Rees, D. (2024). *The impact of artificial intelligence on output and inflation* (Working Paper No. 1179). Bank for International Settlements.

Aldasoro, I., Gambacorta, L., Korinek, A., Shreeti, V., & Stein, M. (2024). *Intelligent financial system: How AI is transforming finance* (Working Paper No. 1194). Bank for International Settlements.

Autor, D. H. (2024). *Applying AI to rebuild middle class jobs* (Working Paper No. 32140). National Bureau of Economic Research.

Bank for International Settlements. (2024). Artificial intelligence and the economy: Implications for central banks. In *Annual economic report 2024* (chap. III). Basel: Bank for International Settlements.

Bonfiglioli, A., Crinò, R., Filomena, M., & Gancia, G. (2025). *Data, power and emissions: The environmental cost of AI* (Discussion Paper No. 20686). Centre for Economic Policy Research.

Brynjolfsson, E., Korinek, A., & Agrawal, A. K. (2025). *A research agenda for the economics of transformative AI* (Working Paper No. 34256). National Bureau of Economic Research.

Brynjolfsson, E., Li, D., & Raymond, L. R. (2025). Generative AI at work. *Quarterly Journal of Economics*, *140*(2), 889–942.

Brynjolfsson, E., Rock, D., & Syverson, C. (2018). Artificial intelligence and the modern productivity paradox: A clash of expectations and statistics. In A. Agrawal, J. Gans,

& A. Goldfarb (Eds.), *The economics of artificial intelligence: An agenda.* Chicago: University of Chicago Press.

Brynjolfsson, E., Rock, D., & Syverson, C. (2021). The productivity J-curve: How intangibles complement general purpose technologies. *American Economic Journal: Macroeconomics*, *13*(1), 333–372.

Chi, T.-C., Fan, T.-H., Ghigliazza, R., Giannone, D., & Wang, Z. K. (2025). *Macroeconomic forecasting and machine learning* (Discussion Paper No. 20727). Centre for Economic Policy Research.

Dell, M. (2025). Deep learning for economists. *Journal of Economic Literature*, *63*(1), 5–58.

Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science*, *384*(6702), 1306–1308.

Felten, E. W., Raj, M., & Seamans, R. (2023). *Occupational heterogeneity in exposure to generative AI* (Working Paper). SSRN.

Fernández-Villaverde, J. (2025). *Deep learning for solving economic models* (Discussion Paper No. 20669). Centre for Economic Policy Research.

Financial Stability Board. (2024, November). *The financial stability implications of artificial intelligence* (Tech. Rep.). Financial Stability Board.

Gambacorta, L., Kharroubi, E., Mehrotra, A., & Oliviero, T. (2026). *Artificial intelligence and growth in advanced and emerging economies: Short-run impact* (Discussion Paper No. 20992). Centre for Economic Policy Research.

International Monetary Fund. (2024, October). Advances in artificial intelligence: Implications for capital market activities. In *Global financial stability report* (chap. 3). Washington, DC: International Monetary Fund.

Korinek, A. (2023). Generative AI for economic research: Use cases and implications for economists. *Journal of Economic Literature*, *61*(4), 1281–1317.

Korinek, A. (2024). *Economic policy challenges for the age of AI* (Discussion Paper No. 19539). Centre for Economic Policy Research.

Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, *381*(6654), 187–192.

Stanford Institute for Human-Centered Artificial Intelligence. (2025). *Artificial intelligence index report 2025* (Tech. Rep.). Stanford University.

Thuemmel, U. (2023). Optimal taxation of robots. *Journal of the European Economic Association*, *21*(3), 1154–1190.